Psi4 for OSPO Internship

Lori Burns, Sherrill Group, 17 March 2026

QUANTUM CHEMISTRY

applying the laws of physics to molecular systems

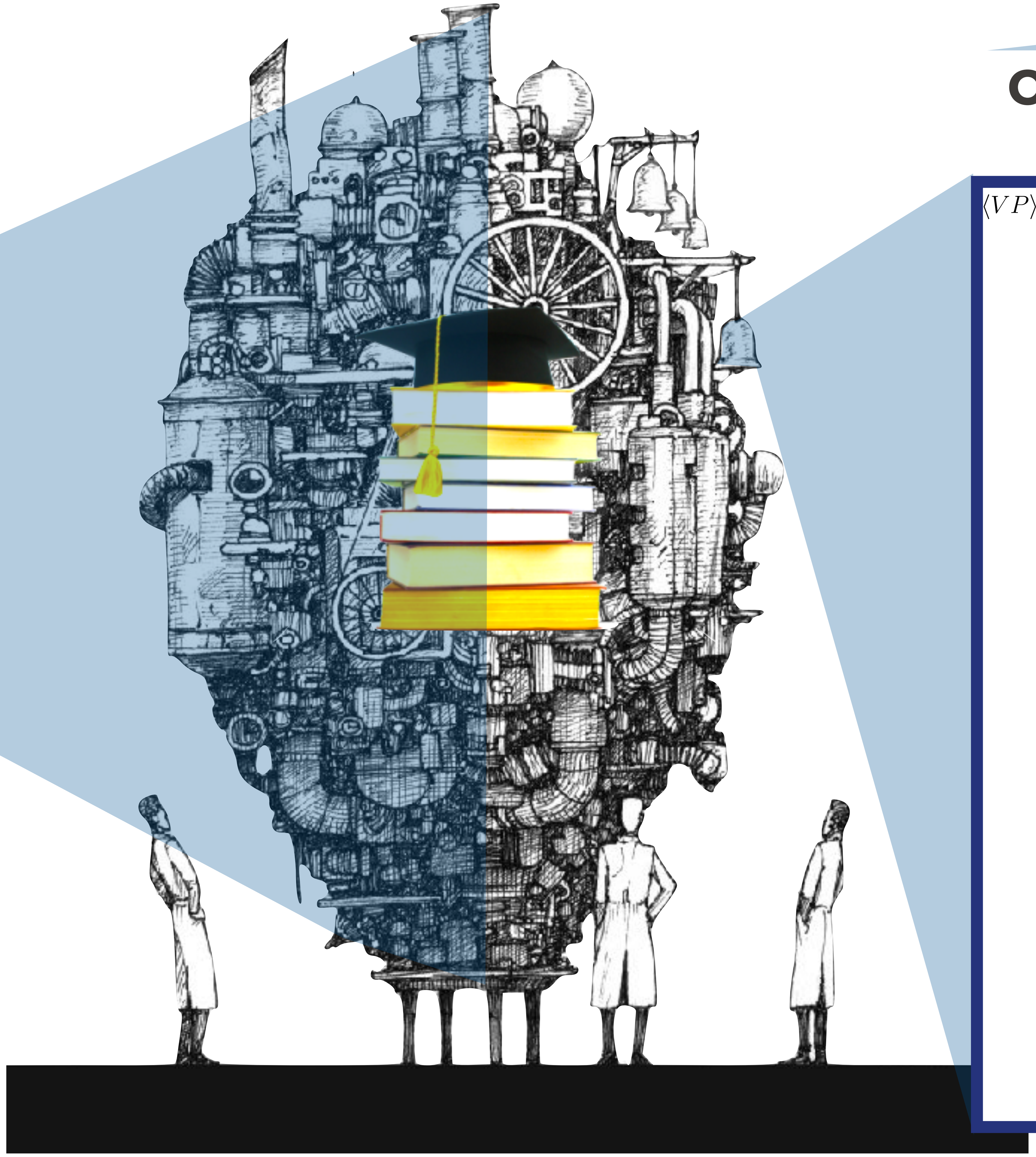
WORLDWIDE QC PROGRAMS

- adcc
- cfour
- gamess
- entos
- mrchem
- qchem
- torchani
- terachem
- terachem_pbs
- nwchem
- xtb
- turbomole
- mace
- molpro
- psi4
- s-dftd3
- dftd3
- mopac
- aimnet2
- dftd4
- rdkit
- openmm
- mp2d
- mctc-gcp
- gcp

- **METHOD ZOO** hundreds of methods with different performance on different local chemical domains.
- **PHYSICS** little to exact. **SCALE** N^3 to N^8
- QC in **DEMAND** for drug design, materials design, quantum compuing, training cheaper methods, ML, & AI

A QC PROGRAM

(strong coupling, low cohesion)



SCHRÖDINGER EQ'N, 1926

$$\hat{H}\Psi = E\Psi$$

$$\hat{H} = \underbrace{-\sum_{A}^{\text{nuc}} \frac{1}{2M_A}\nabla_A^2}_{\text{nuclear kinetic energy}} \underbrace{-\frac{1}{2}\sum_{i}^{\text{elec}}\nabla_i^2}_{\text{electronic kinetic energy}} \underbrace{-\sum_{A}^{\text{nuc}}\sum_{i}^{\text{elec}}\frac{Z_A}{r_{Ai}}}_{\text{electron/nuclear attraction}} + \underbrace{\sum_{A>B}^{\text{nuc}}\frac{Z_A Z_B}{R_{AB}}}_{\text{nuclear/nuclear repulsion}} + \underbrace{\sum_{i>j}^{\text{elec}}\frac{1}{r_{ij}}}_{\text{electron/electron repulsion}}$$

ONE TERM OF ONE METHOD

(those are tensors)

$$\begin{aligned}
\langle VP\rangle = {} & -4.0\tilde{v}^{i'i}_{i'j}S_{ij} - 8\tilde{v}^{i'j'}_{i'j'}S_{ij}S_{ij} + 4\tilde{v}^{i'j}_{i'j'}S_{ij}S_{ij'} + 2\tilde{v}^{ii}_{i'j}S_{i'j} + 4\tilde{v}^{ij'}_{i'j'}S_{i'j}S_{ij} \\
& - 2\tilde{v}^{ij}_{i'j'}S_{i'j}S_{ij'} - 2\tilde{v}^{ii}_{jj} - 4\tilde{v}^{ij'}_{jj'}S_{ij} + 2\tilde{v}^{ij}_{jj'}S_{ij'} - 2\gamma^{B}_{uu'}\tilde{v}^{i'i}_{i'u'}S_{iu} \\
& - 4\gamma^{B}_{uu'}\tilde{v}^{i'j}_{i'j}S_{iu}S_{iu'} + 2\gamma^{B}_{uu'}\tilde{v}^{i'j}_{i'u'}S_{ij}S_{iu} + 2\gamma^{B}_{uu'}\tilde{v}^{i'j}_{i'u}S_{ij}S_{iu'} - 4\gamma^{B}_{uu'}\tilde{v}^{i'u}_{i'u'}S_{ij}S_{ij} \\
& + \gamma^{B}_{uu'}\tilde{v}^{ii}_{i'u'}S_{i'u} + 2\gamma^{B}_{uu'}\tilde{v}^{ij}_{i'j}S_{i'u}S_{iu'} - \gamma^{B}_{uu'}\tilde{v}^{ij}_{i'u'}S_{i'u}S_{ij} - \gamma^{B}_{uu'}\tilde{v}^{ij}_{i'u}S_{i'j}S_{iu'} \\
& + 2\gamma^{B}_{uu'}\tilde{v}^{iu}_{i'u'}S_{i'j}S_{ij} + \gamma^{B}_{uu'}\tilde{v}^{ij}_{ju}S_{iu'} - 2\gamma^{B}_{uu'}\tilde{v}^{iu}_{ju'}S_{ij} - \gamma^{B}_{uu'}\tilde{v}^{ii}_{uu'} - 2\gamma^{B}_{uu'}\tilde{v}^{ij}_{uj}S_{iu'} \\
& + \gamma^{B}_{uu'}\tilde{v}^{ij}_{uu'}S_{ij} - 2\Gamma^{B}_{uu'u''u'''}\tilde{v}^{i'u}_{i'u'}S_{iu''}S_{iu'''} + \Gamma^{B}_{uu'u''u'''}\tilde{v}^{iu}_{i'u'}S_{i'u''}S_{iu'''} \\
& - \Gamma^{B}_{uu'u''u'''}\tilde{v}^{iu}_{u''u'}S_{iu'''} - 4\gamma^{A\alpha}_{tt'}\gamma^{B\alpha}_{uu'}\tilde{v}^{ij}_{ij}S_{t'u}S_{tu'} + 2\gamma^{A\alpha}_{tt'}\gamma^{B\alpha}_{uu'}\tilde{v}^{ij}_{iu'}S_{t'u}S_{tj} \\
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& + \gamma^{A\alpha}_{tt'}\gamma^{B\alpha}_{uu'}\tilde{v}^{t'j}_{uu'}S_{tj} - \gamma^{A\alpha}_{tt'}\gamma^{B\alpha}_{uu'}\tilde{v}^{t't}_{uu'} - 2\gamma^{A\alpha}_{tt'}\Gamma^{B\alpha\alpha+\beta\alpha}_{uu'u''u'''}\tilde{v}^{iu}_{iu'}S_{t'u''}S_{tu'''} \\
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\end{aligned}$$

image from https://patientsafe.wordpress.com/2016/05/29/the-healthcare-safety-machine-is-broken/

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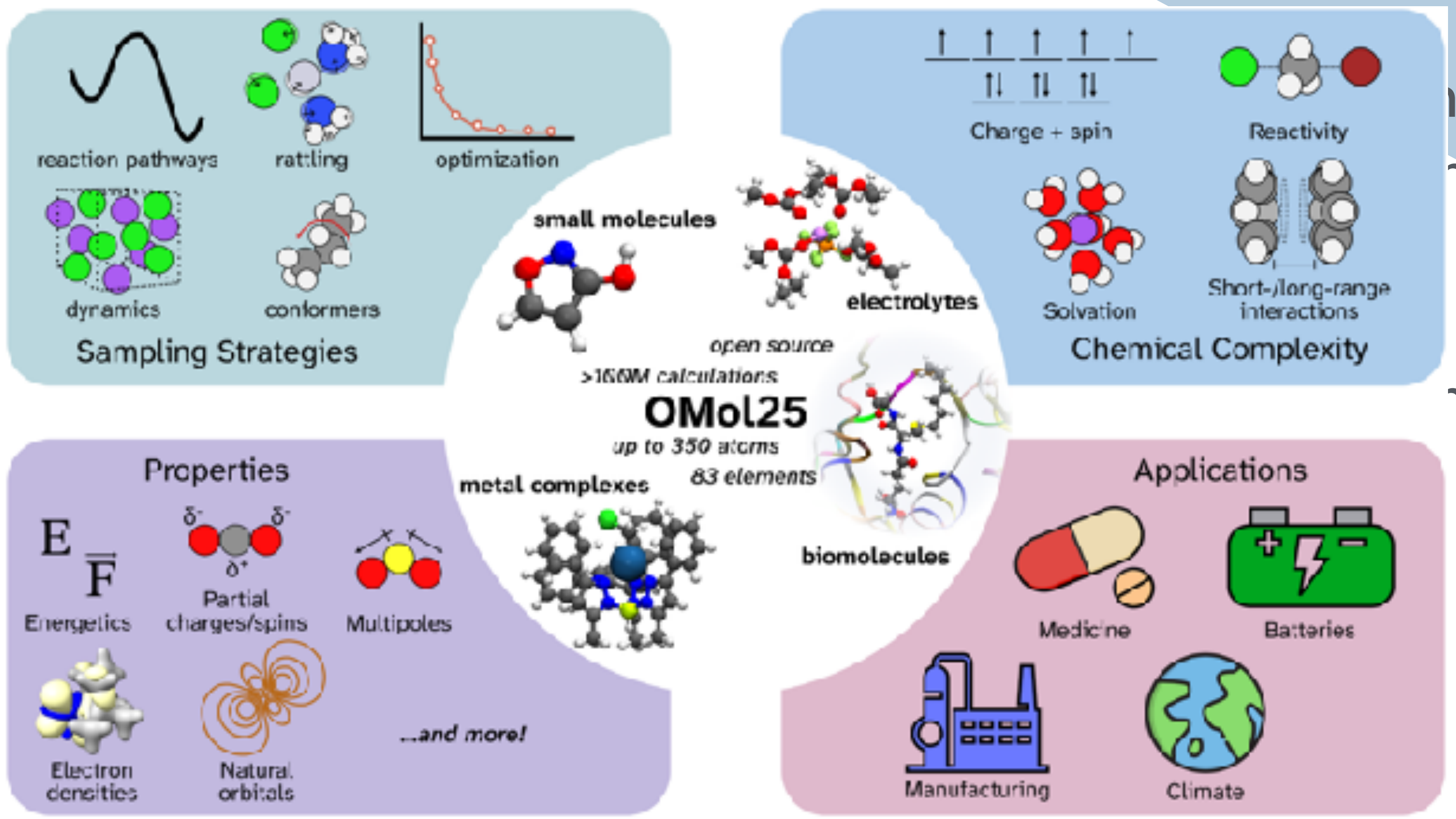
- mctc-gcp
- gcp



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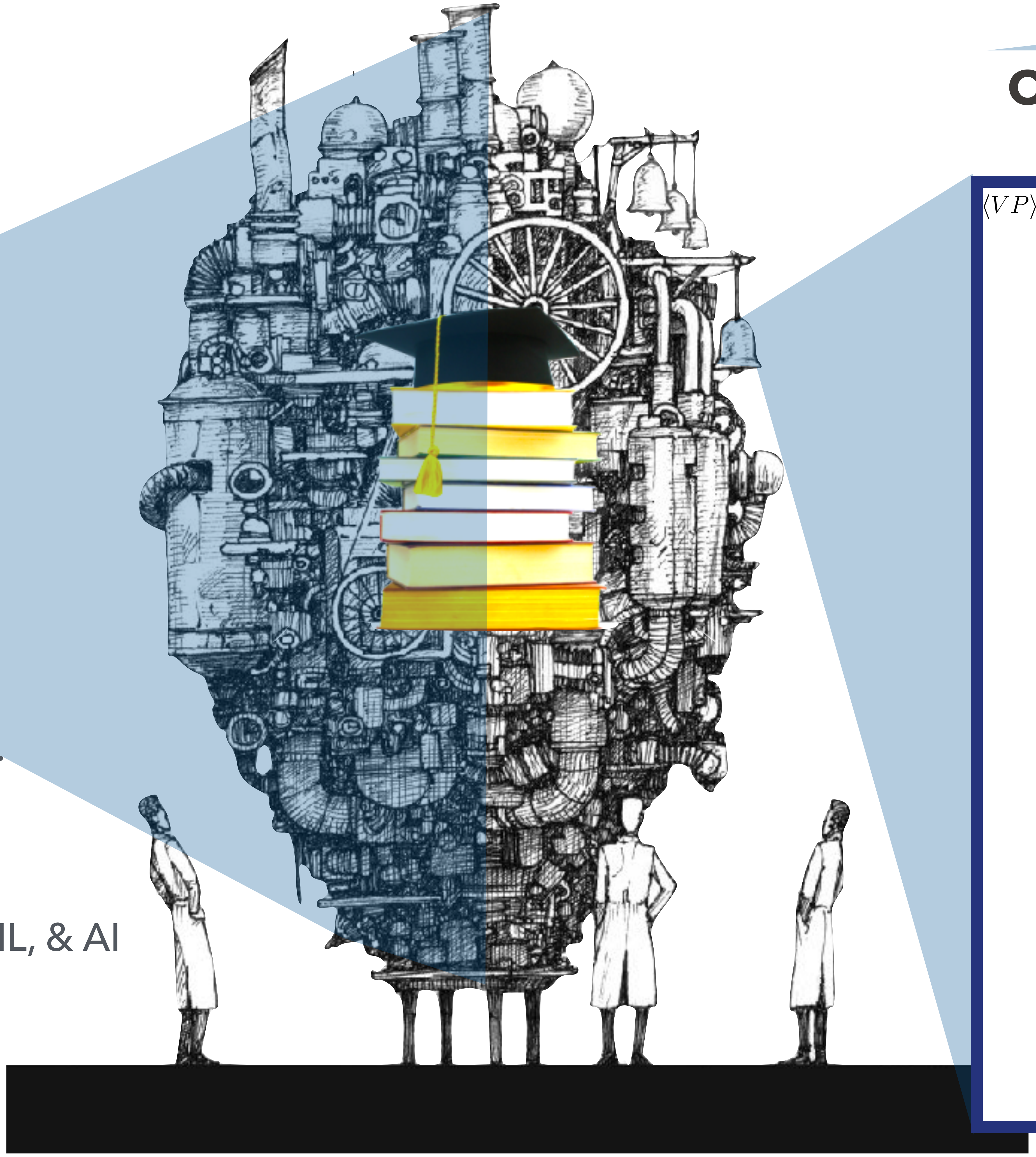

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\end{aligned}$$

image from https://patientsafe.wordpress.com/2016/05/29/the-healthcare-safety-machine-is-broken/

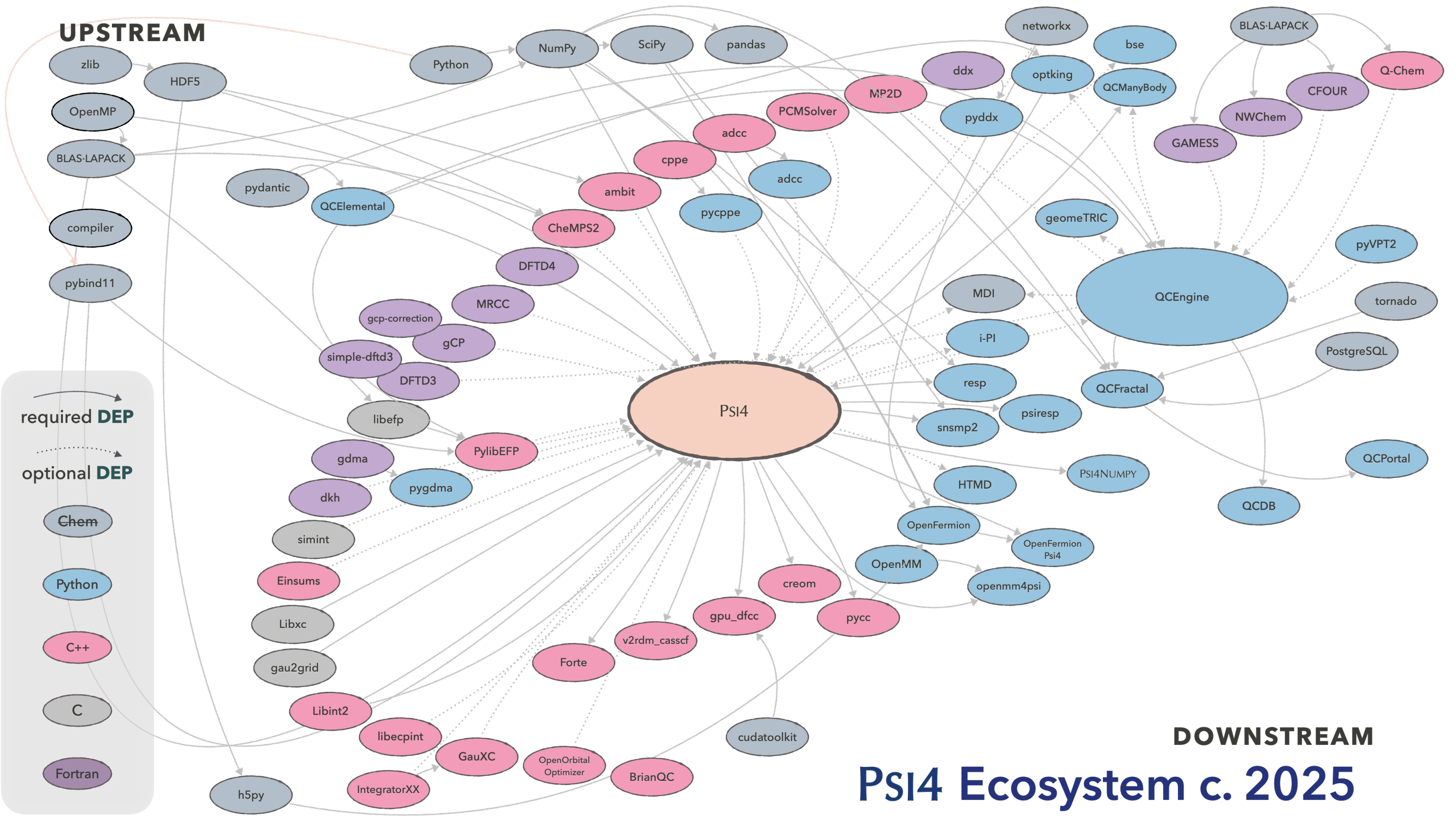
```
import psi4

psi4.geometry('''
Ne
Ne 1 3.0
''')

psi4.set_options({
 'freeze_core': 'True'})

psi4.energy('ccsd(t)/cc-pvtz')
```

- **SMP** parallelism
- **70/30%** C++/Python for easy production-quality user access
- **PY/C++** interface layer below the user layer makes it simple to optimize a reference implementation into a compiled language.

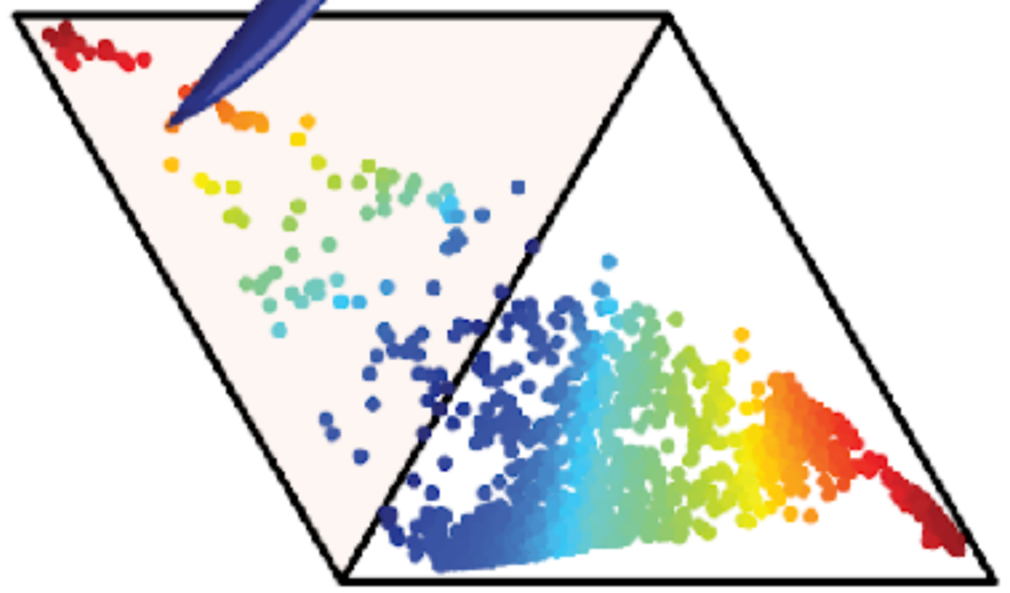


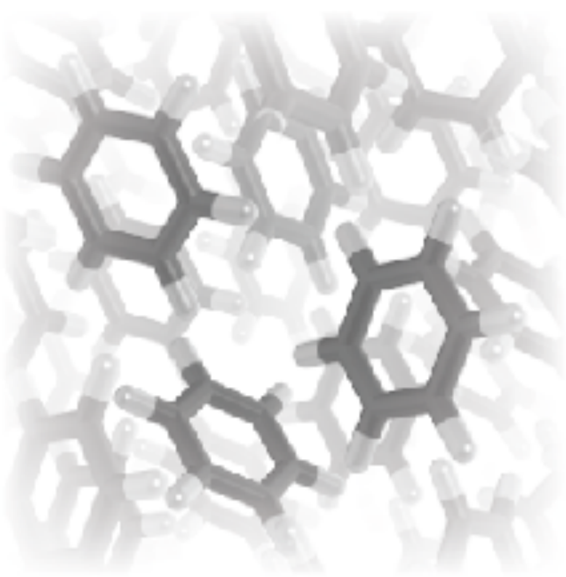
Psi4 Ecosystem c. 2025

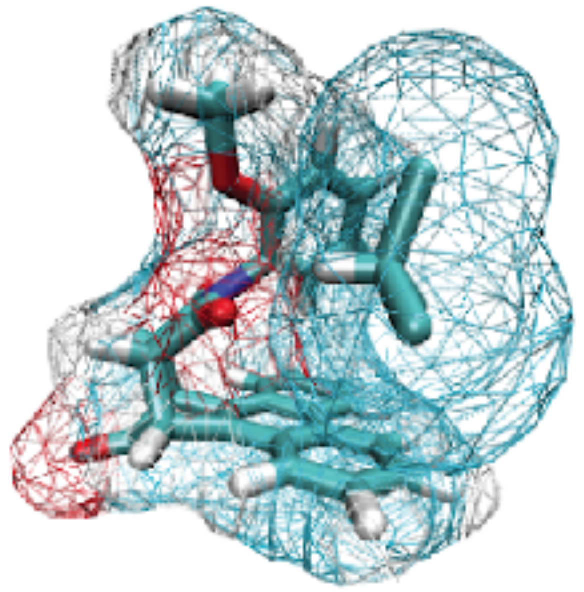
Library API • C++/Python Interface • External Ecosystem

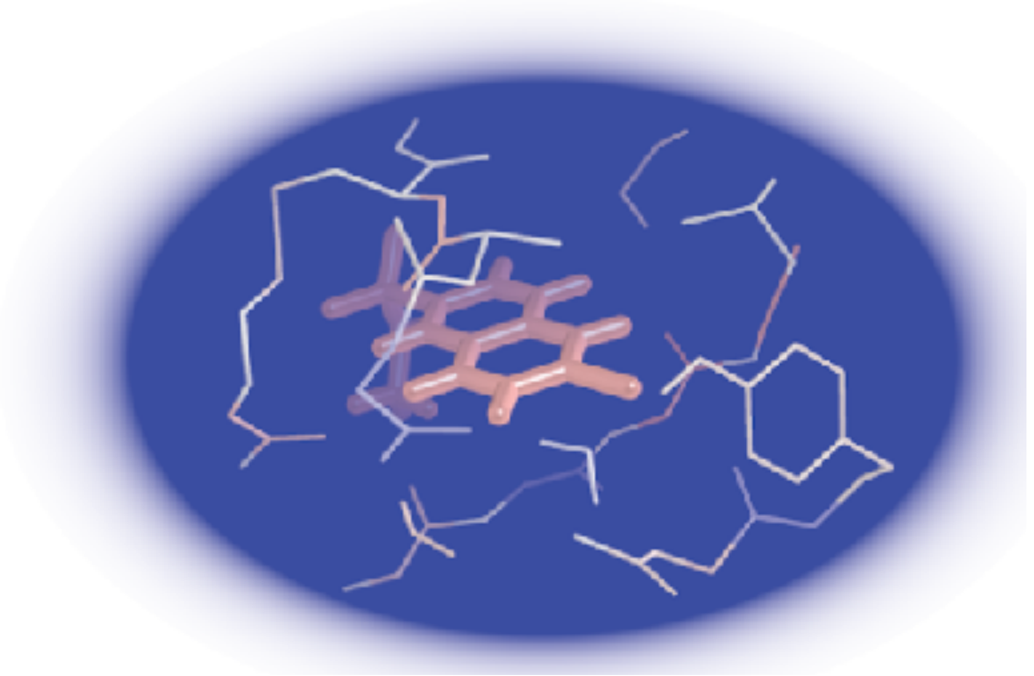
Psi4

OPEN-SOURCE QUANTUM CHEMISTRY









Psi4 Stars 1.1k

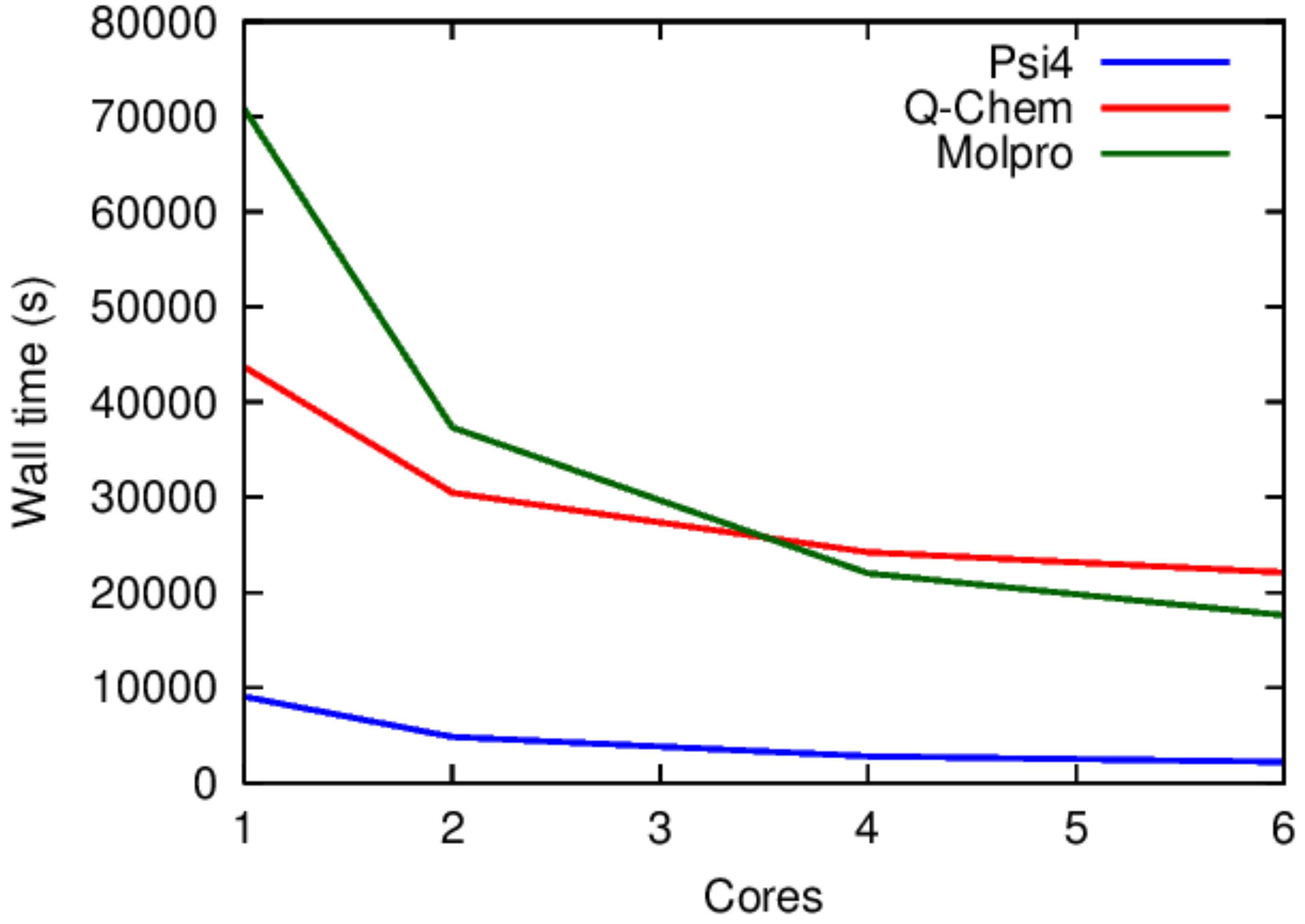
conda-forge Psi4 downloads 359k

a QC Program (developed at GT) user

PROJECTS

no quantum chemistry required

A: PERFORMANCE OPTIMIZATION



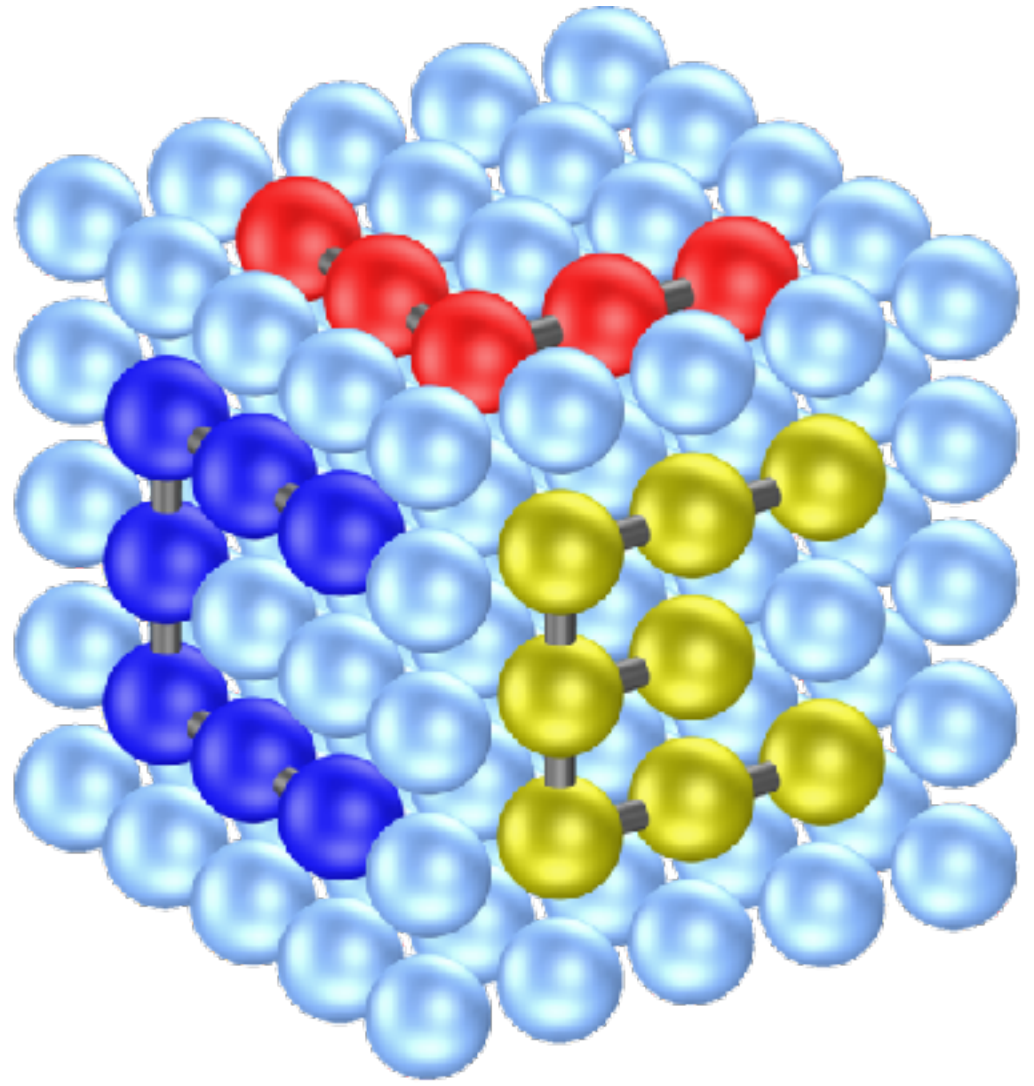
- For many types of jobs, we are competitive with or better than commercial codes.
- After recent work, some QC-intensive portions of the code have been much accelerated.
- Looking for help beating down the remaining time requirements for less QC portions.
- Profiling, OpenMP, and light C++

B: PERFORMANCE REGRESSION TESTING

```
          C_DCOPY(nocc, Cp[i], 1, Coccp[i], 1);
      }
-     // Scale by occ
+     // Scale by sqrt(occ)
      for (int i = 0; i < nocc; i++) {
-         C_DSCAL(nbf, occ->get(i), &Coccp[0][i], nocc);
+         C_DSCAL(nbf, std::sqrt(occ->get(i)), &Coccp[0][i], nocc);
      }
      // Form D = Cocc*Cocc'
      D->gemm(false, true, 1.0, Cocc, Cocc, 0.0);
```

- In a recent release, we made a correct adjustment, and it passed all 7k tests. But it turns out to significantly impair convergence.
- We'd like help setting up a performance regression dashboard. We already have the start of a suite of production-length calculations.

C: CRYS2LATTE



CRYSTALATTE

- Separate codebase for high-performance calculations on crystals being rewritten this summer with symmetry imposed.
- We'd like help with performance optimizations.

***7713** total tests in suite*

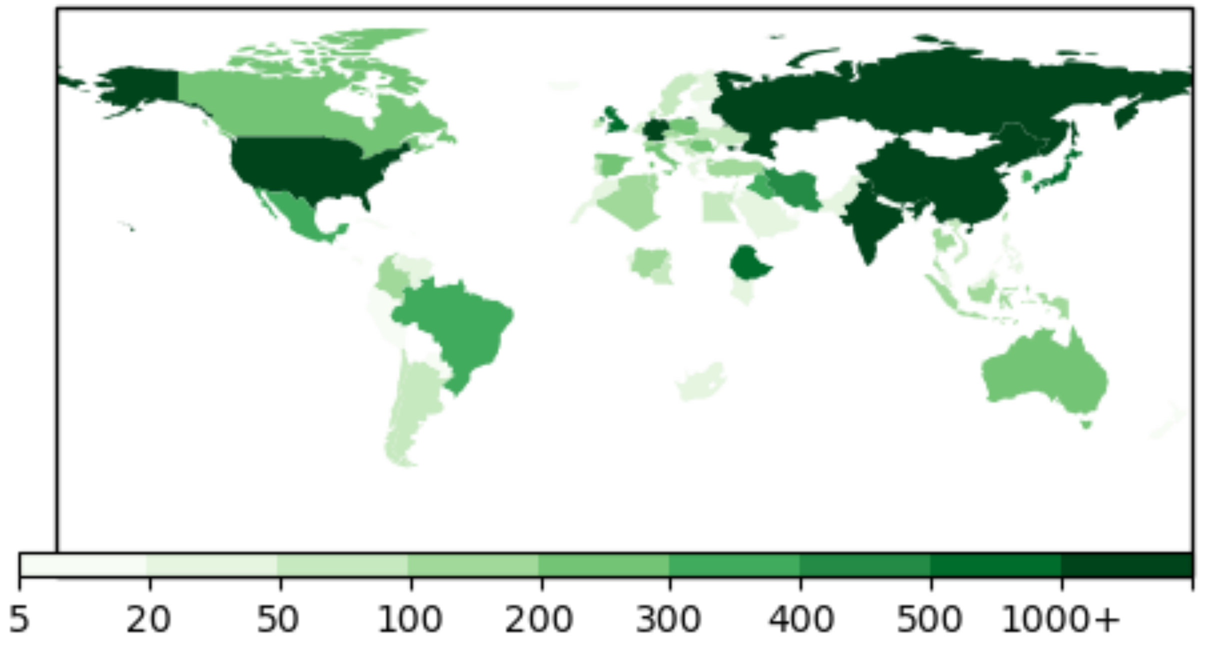
build passing

CONTINUOUS INTEGRATION TESTING

a repository's lifeline and anchor

OPEN DEVELOPMENT

- Over 15000 commits
- 119 contributors
- 1100 GitHub stars
- GitHub code review
- Automatic checks through continuous integration services
- LGPL license



PSI4 INTERNAL CI, AZURE

__2296__ tests of internal Psi4 capabilities, Azure is cmdline-like for users, non-conda compilers

psi4 / psi4 / Pipelines / psi4.psi4 / 20250618.3

Name	Status	Duration	
Windows VS2022	Success	1h 43m 18s	quick
Linux Builds gcc_11	Success	1h 5m 50s	quick + 1/3 full, earliest GCC, Py
Linux Builds gcc_latest	Success	1h 57m 12s	quick + 1/3 full, latest GCC, Py
Linux Builds clang_latest	Success	1h 19m 26s	quick + 1/3 full

PSI4 ECOSYSTEM CI, GHA

__<386__ tests integrating with external software, architecture variety, all-conda

Docs / Propose Docs (ubuntu-latest, 3.10) (pull_request)	Successful in 28m	
Eco / Eco • 3.10 • Conda Clang (M-Silicon) (pull_request)	Successful in 37m	addon + quick
Eco / Eco • 3.11 • Conda Clang (M-Intel) (pull_request)	Successful in 103m	addon + quick
Eco / Eco • 3.12 • LLVM clang-cl (W) (pull_request)	Successful in 60m	addon
Eco / Eco • 3.13 • Conda GNU (L) (pull_request)	Successful in 45m	addon
psi4.psi4 — #20250618.3 succeeded	Required	